Optimal Reconfiguration of Systems in Aeronautics

Lecture 10: "Exact Model Matching"

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Outline

1. Nominal and Post-Fault Exact Model Matching
2. B747 Application Example

Objective I

This lecture provides a fault tolerant perspective on model matching.

It shows how a classical assignment scheme based on the pseudo-inverse matrix may be adapted by making use of the modified pseudo-inverse technique.

Outline

1. Nominal and Post-Fault Exact Model Matching
   - Introduction
   - Nominal Operation
   - Faulty Operation
2. B747 Application Example

Preliminaries I

Properties of interest in the analysis of the nominal and post-fault systems range from stability to optimality. Analyzing stability for linear systems is done using the eigenvalues from the spectrum of the closed-loop system, which determine the ideal operation when being placed in certain desirable regions of the complex-plane (for the complete characterization of the system behavior, the eigenvectors should however be considered, as well).

After faults impair the nominal system, Exact Model Matching is classically used to recover the same performance also in the post-fault operation. Unfortunately, this approach usually conducts to unrealistic control gains or there is no solution at all most often and therefore approximate solutions based on the Pseudo-Inverse Method might be preferred, which also encounters instability problems. These are solved however by its extension called the Modified Pseudo-Inverse Method; again, the latter escapes stability for certain fault conditions.

Evidently, in all these cases, stability represents the critical issue to be solved, but important are also the computational efficiency in real-time accommodation of the faulty system altogether with the realistic treatment of robustness. Among the appealing counteractions, Pareto-optimality seems to be appropriate for the characterization of passive robustness, while the robust optimality appears to be useful for its active treatment.

Preliminaries II

To this end, two model matching techniques for the accommodation of the post-fault controller are approached, namely the Admissible Model Matching and the Robust Optimal Model Matching, the first proposing to search in the class of admissible models not the best one but any model which provides a stable admissible behavior at the moment of interest, while the second suggests redetermining the weighting symmetric matrices of a post-fault quadratic integral in order to guarantee the optimal matching of the two closed-loop systems from the nominal and faulty conditions.

The benefits brought by using the model matching post-fault accommodation are illustrated along with the development of an application to the short-period mode control of the longitudinal motion of an aircraft.

Nominal operation I

The Exact Model Matching (EMM) technique with State-Feedback (SF) aims at designing the linear control law

$$u_n(t) = -K_n x(t)\,, \tag{1}$$

for the continuous linear time-invariant (LTI) system whose nominal operation is modeled by

$$M_n \;:\; \dot{x}(t) = A_n x(t) + B_n u_n(t)\,, \tag{2}$$

such that the resulting closed-loop behavior matches that of the reference model

$$\hat{M}_n^* \;:\; \dot{x}(t) = \hat{A}_n^* x(t)\,, \tag{3}$$

where $A_n \in \mathbb{R}^{n\times n}$ and $B_n \in \mathbb{R}^{n\times m}$ are the nominal system and control matrices, $x(t) \in \mathbb{R}^n$ and $u(t) \in \mathbb{R}^m$ are the state and control vectors, $\hat{A}_n^* \in \mathbb{R}^{n\times n}$ is a given stable matrix, and $K_n \in \mathbb{R}^{m\times n}$ is the state-feedback matrix that has to be determined.

The solution of the exact model matching problem is obtained by solving the matrix equation

$$\hat{A}_n \triangleq A_n - B_n K_n = \hat{A}_n^*\,. \tag{4}$$

Nominal operation II

In this respect, the necessary and sufficient Erzberger condition for a solution to exist is

$$\mathrm{Im}\left(A_n - \hat{A}_n^*\right) \subseteq \mathrm{Im}\,(B_n)\,, \tag{5}$$

and the solution is given by

$$K_n = B_n^\dagger (A_n - \hat{A}_n^*)\,, \tag{6}$$

which provides the basis for the Pseudo-Inverse Method (PIM); note that the pseudo-inverse matrix is used in theory, but, in practice, linear systems are usually solved in a least squares sense.

In (6), notation $(.)^\dagger$ stands for the pseudo-inverse matrix, whose computation technique is in order. If matrix B_n is with $\mathrm{rank}\, B_n = r \leq m$, its pseudo-inverse $B_n^\dagger$ is given by the following theorem.

Nominal operation III

Theorem (Singular Value Decomposition and Pseudo-Inverse Computation)

Given the matrix $B_n \in \mathbb{R}^{n \times m}$, *with rank* $r \leq m$, *there exist orthogonal matrices*

$$U_n = \begin{bmatrix} u_1 & \dots & u_n \end{bmatrix} \in \mathbb{R}^{n \times n} \text{ and } V_n = \begin{bmatrix} v_1 & \dots & v_m \end{bmatrix} \in \mathbb{R}^{m \times m},$$

such that

$$S_n = U_n^\top B_n V_n = \begin{bmatrix} \Sigma_n & 0 \\ 0 & 0 \end{bmatrix},$$

for

$$\Sigma_n = \begin{bmatrix} \sigma_{n_1} & & \\ & \ddots & \\ & & \sigma_{n_r} \end{bmatrix} \in \mathbb{R}^{r \times r},$$

where

$$\sigma_{n_1} \geq \sigma_{n_2} \geq \dots \geq \sigma_{n_r} > 0\,.$$

Nominal operation IV

Theorem (Singular Value Decomposition and Pseudo-Inverse Computation – cont'd)

Moreover, the pseudo-inverse of B_n has the form

$$B_n^\dagger = V_n \begin{bmatrix} \Sigma_n^{-1} & 0 \\ 0 & 0 \end{bmatrix} U_n^\top \; ;$$

vectors u_{n_i} and v_{n_i} are the i-th left- and right- singular vectors, respectively, and σ_{n_i} are the singular values of B_n. $\square$

Nominal operation V

However, if the linear reference model M_n^* is given directly through its open-loop state equation

$$M_n^* \; : \; \dot{x}_m(t) = A_m x_m(t) + B_m u_m(t) \,, \tag{7}$$

then the exact model matching turns out to be alike the Perfect Model Following (PMF) problem, which means to find the control gain that cancels the error between the states of the two systems, (2) and (7), respectively

$$e(t) = x(t) - x_m(t) \,,$$

for the whole time interval, i.e., $e(t) \equiv 0$, $\forall\, t \geq 0$, when starting from the same initial condition and submitted to the same control input.

In this situation, PIM is a special case of the implicit Linear Model Following (LMF), as the plant and reference systems are replaced by the nominal and faulty models of the same plant.

Faulty operation I

Whether parametric faults are considered, assuming that an LTI model is still able to capture its behavior, the system to be controlled is described by the pair (A_f, B_f), that is the post-fault system operation is modeled by

$$M_f : \ \dot{x}(t) = A_f x(t) + B_f u_f(t) \,, \tag{8}$$

and the SF-EMM technique aims at designing the linear control law

$$u_f(t) = -K_f x(t) \,, \tag{9}$$

such that the resulting closed-loop behavior matches that of the reference model from (3).

As before, the solution of the exact model matching problem in the faulty situation amounts to solving the matrix equation

$$\hat{M}_f : \ \hat{A}_f \triangleq A_f - B_f K_f = \hat{A}_n^* \,. \tag{10}$$

Faulty operation II

Similarly to the nominal situation, the necessary and sufficient Erzberger condition for a solution to exist is

$$\mathrm{Im}\left(A_f - \hat{A}_n^*\right) \subseteq \mathrm{Im}\,(B_f)\,, \tag{11}$$

and the solution is given by

$$(K_f^{PIM} \equiv)\; K_f = B_f^{\dagger}(A_f - \hat{A}_n^*)\,. \tag{12}$$

This far, (1)-(6) and (8)-(12) define the Exact Model Matching (EMM) problems in the nominal and post-fault system operations.

At any rate, condition (5) is obviously satisfied from the system nominal design in the healthy situation but its faulty counterpart (11) will hold only for very particular faults and therefore no exact solution will exist in most fault cases.

For this reason, approximate rather than exact solutions may be of interest.

Outline

1. Nominal and Post-Fault Exact Model Matching
2. B747 Application Example
   - Flight Condition
   - Nominal System
   - Faulty System

Flight condition I

The aircraft flight condition is assumed to be a straight and constant level flight at fixed altitude of 40000 *ft* and Mach number 0.8.

Considering the CoG (center of gravity) position to be $(0.25, 0, 0)$ of m.a.c. (mean aerodynamic chord) and the aircraft mass 19792 *slug*, the corresponding steady-state speed is 774 *fps*.

The control δ_e (*rad*) is the elevator deflection.

Nominal system I

Remember that the short-period motion involves rapid changes in the angle-of-attack α (rad) and pitch angle θ (rad) at roughly constant speed, with the specification that, in the small-disturbance case analysis, the evolution of α and w follow approximately the same shape.

Open-loop model I

The aircraft longitudinal short-period (SP) dynamics is described by the state vector

$$x = \begin{bmatrix} w & q \end{bmatrix}^\top ,$$

where w (ft/sec) represents the inertial velocity in the z-direction of F_B and q (rad/sec) represents the pitch rate (F_B denotes the aircraft body-axis reference frame).

The linearized model of the system is

$$\dot{x}(t) = A_n x(t) + B_n u_n(t) ,$$

where the system and control matrices are given as

$$A_n = \begin{bmatrix} -0.3127 & 774.0000 \\ -0.0010 & -0.4284 \end{bmatrix} \text{ and } B_n = \begin{bmatrix} -17.9416 \\ -1.1577 \end{bmatrix} .$$

Open-loop model II

In this flight configuration, matrix A_n has eigenvalues

$$\Lambda\left(A_n\right) = \left\{-0.3706 \pm 0.8779\jmath\right\}.$$

The name "short-period" obviously follows from the aircraft natural mode having a period of 7.08 *sec*. It can also be observed that it is heavily damped, from the magnitude of the damping coefficient which is 0.389.

Exact model matching I

Although the B747 short-period natural mode is inherently stable, the handling qualities of the system may be enhanced by choosing an appropriate closed-loop reference model and using an exact model matching controller design.

The handling qualities refer to those characteristics of an aircraft that govern the ease and precision with which a pilot is able to perform his tasks. They can be expressed in terms of the damping coefficient and natural frequency of the system mode, as these two parameters govern the system dynamic stability.

Exact model matching II

In this respect, it can then be checked that the reference model

$$\hat{A}_n^* = \begin{bmatrix} -0.4388 & 726.7417 \\ -0.0091 & -3.4778 \end{bmatrix}, \tag{13}$$

with eigenvalues

$$\Lambda\left(\hat{A}_n^*\right) = \left\{\hat{\lambda}_{n_1}^*, \hat{\lambda}_{n_2}^*\right\} = \{-1.9583 \pm 2.0813\jmath\},$$

provides satisfactory performance, as far as longitudinal short-period oscillation is concerned; indeed, the reference model (13) has a damping coefficient of about 0.6 (0.685) and a natural frequency of approximately 3 (2.85) *rad/sec*, that the classical "thumb-print" criterion recommends as a good target for the aircraft short-period mode performance.

Exact model matching III

The state-feedback controller

$$u_n(t) = -K_n x(t)\,,$$

with

$$K_n = \begin{bmatrix} -0.0070 & -2.6340 \end{bmatrix}\,, \tag{14}$$

achieves exact model matching and it is therefore chosen as the nominal system controller.

Considered faults I

In what follows, the considered faults are the loss-of-effectiveness of the actuator that controls the elevator deflection. Such faults are classically modeled by a multiplicative factor $\tau_f \in (0, 1]$, so that the linear state-space representation of the faulty system will be described by the matrices $A_f = A_n$ and $B_f = (1 - \tau_f) B_n$.

Obviously, $\tau_f = 0$ represents the nominal operation case and $\tau_f = 1$ stands for the complete failure case, which is the situation when the actuator does not act on the system anymore.

The solution to the exact model matching problem is naturally obtained when replacing the nominal pair (A_n, B_n) in (6) with the faulty pair (A_f, B_f).

The control-gain matrix becomes

$$K_f = \frac{1}{1 - \tau_f} B_n^{\dagger} (A_n - \hat{A}_n^*) \,, \tag{15}$$

and it then follows that all faults are recoverable (except obviously for $\tau_f = 1$, since the system is no longer controlled at all in that case).

Considered faults II

Numerical values of K_f are given in the following table, for increasing values of τ_f.

τ_f	0.1	0.2	0.3	0.4
k_{f_1}	-0.0078	-0.0088	-0.0100	-0.0117
k_{f_2}	-2.9267	-3.2925	-3.7629	-4.3900
ζ_f	0.7126	0.7455	0.7858	0.8365
ω_f	2.9955	3.1593	3.3582	3.6064

τ_f	0.5	0.6	0.7	0.8	0.9
k_{f_1}	-0.0141	-0.0176	-0.0234	-0.0351	-0.0703
k_{f_2}	-5.2680	-6.5850	-8.7800	-13.1700	-26.3400
ζ_f	0.9029	0.9942	1.0000	1.0000	1.0000
ω_f	3.9275	4.3652	3.0237	2.6664	2.4458

Considered faults III

However, such a convenient case is far from being real; first, power limitations and actuator saturations have the consequence that the above solution works only for a limited range of values of τ_f; second, the assumption that, after the fault, the system matrix remains identical with that from the nominal case, i.e., $A_f = A_n$, is not too realistic since A_n results from the linearization of a nonlinear behavior around an operating point that might well be changed by the fault occurrence.

In order to consider more complex situations, it is assumed that the fault is described by two coefficients, i.e., not only the control matrix $B_f = (1 - \tau_f)B_n$ but also the system matrix $A_f = (1 - \gamma_f)A_n$, thus indicating a multiplicative fault, where $\gamma_f \in (0, 0.1]$ is a given factor.

Still, it can be checked that the faulty counterpart of the consistency condition (5) does not hold anymore for $\gamma_f \neq 0$, therefore the faulty counterpart of (6) cannot be used for computing the post-fault control matrix.

Questions? vs. Answers!

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